Performance Analysis of CPU Scheduling Algorithms

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1. Introduction

CPU scheduling is a fundamental operating system function that decides which process runs in the memory at any given time. The efficiency of the CPU depends heavily on the scheduling algorithm used to manage the queue of ready processes.

This report evaluates the performance of **five** distinct scheduling algorithms: **First-Come, First-Served (FCFS)**, **Shortest Job First (SJF) Non-Preemptive, Priority Non-Preemptive**, **Round Robin (RR),** and **Multi-Level Queue (MLQ)**. These algorithms were tested across five scenarios **(A through E),** representing diverse workload patterns, including variations in burst times, arrival intervals, and idle periods. The analysis focuses on key performance metrics: **Average Waiting Time (AWT), Throughput**, and **CPU Utilization**.

2. Scenario-Based Results

Scenario A Results

High utilization workload with varying burst lengths.

FCFS

Gantt: `0-5[P1] 5-8[P2] 8-16[P3] 16-22[P4] 22-24[P5] 24-28[P6] 28-29[P7] 29-34[P8]`

PID	WT	TAT	RT	CT
P1	0.0	5.0	0.0	5.0
P2	4.0	7.0	4.0	8.0
P3	6.0	14.0	6.0	16.0
P4	13.0	19.0	13.0	22.0
P5	18.0	20.0	18.0	24.0
P6	18.0	22.0	18.0	28.0
P7	21.0	22.0	21.0	29.0
P8	20.0	25.0	20.0	34.0
Avg WT: 12.5	**Throughput:** 0.235	**Util:** 100%		

SJF Non-Preemptive

Gantt: `0-5[P1] 5-7[P5] 7-8[P7] 8-11[P2] 11-15[P6] 15-20[P8] 20-26[P4] 26-34[P3]`

PID	WT	TAT	RT	CT
P1	0.0	5.0	0.0	5.0
P5	1.0	3.0	1.0	7.0
P7	0.0	1.0	0.0	8.0
P2	7.0	10.0	7.0	11.0
P6	5.0	9.0	5.0	15.0
P8	6.0	11.0	6.0	20.0
P4	17.0	23.0	17.0	26.0
P3	24.0	32.0	24.0	34.0
Avg WT: 7.5	**Throughput:** 0.235	**Util:** 100%		

Priority Non-Preemptive

Variant 1 Gantt: `0-5[P1] 5-8[P2] 8-10[P5] 10-16[P4] 16-17[P7] 17-22[P8] 22-30[P3] 30-34[P6]`

Variant 2 Gantt: `0-1[P1] 1-4[P2] 4-6[P5] 6-10[P1] 10-16[P4] 16-17[P7] 17-22[P8] 22-30[P3] 30-34[P6]`

Metric	Variant 1	Variant 2
Avg Waiting Time	9.5	9.125
Throughput	0.235	0.235
Utilization	100%	100%

Round Robin (q=4.0)

Gantt: `0-4[P1] 4-7[P2] 7-11[P3] 11-15[P4] 15-17[P5] 17-18[P1] 18-22[P6] 22-23[P7] 23-27[P8] 27-31[P3] 31-33[P4] 33-34[P8]`

- **Avg Waiting Time:** 14.875
- **Throughput:** 0.235
- **Utilization:** 100%

Multi-Level Queue (MLQ)

- **Queue 1 (RR q=2):** Avg WT: 6.83 | Throughput: 0.272 | Util: 100%
- **Queue 2 (FCFS):** Avg WT: 2.0 | Throughput: 0.142 | Util: 85.7%
- **Queue 3 (Priority):** Avg WT: 14.5 | Throughput: 0.235 | Util: 100%

Scenario B Results

Sparse workload characterized by frequent idle times.

General Note: FCFS, SJF, Priority, and Round Robin produced identical statistics due to the distinct arrival times creating a convoy effect.

FCFS / SJF / Priority / RR (q=4.0)

Gantt: `0-3[P1] 3-5[IDLE] 5-9[P2] 9-10[P3] 10-11[IDLE] 11-13[P4] 13-16[P5] 16-18[P6] 18-20[IDLE] 20-21[P7] 21-25[IDLE] 25-29[P8]`

PID	WT	TAT	RT	CT
P1	0.0	3.0	0.0	3.0
P2	0.0	4.0	0.0	9.0
P3	2.0	3.0	2.0	10.0
P4	0.0	2.0	0.0	13.0
P5	0.0	3.0	0.0	16.0
P6	0.0	2.0	0.0	18.0
P7	0.0	1.0	0.0	21.0
P8	0.0	4.0	0.0	29.0
Avg WT: 0.25	**Throughput:** 0.276	**Util:** 68.97%		

Multi-Level Queue (MLQ)

- **Queue 1 (RR q=2):** Avg WT: 0.0 | Throughput: 0.207 | Util: 58.6%
- **Queue 2 (FCFS):** Avg WT: 0.0 | Throughput: 0.111 | Util: 16.7%
- **Queue 3 (Priority):** Avg WT: 0.125 | Throughput: 0.276 | Util: 68.97%

Scenario C Results

Medium density workload.

FCFS & Priority

Gantt: `0-3[P1] 3-7[P2] 7-9[P3] 9-10[P4] 10-15[P5] 15-17[P6] 17-20[P7] 20-21[P8]`

PID	WT	TAT	RT	CT
P1	0.0	3.0	0.0	3.0
P2	2.0	6.0	2.0	7.0
P3	5.0	7.0	5.0	9.0
P4	6.0	7.0	6.0	10.0
P5	6.0	11.0	6.0	15.0
P6	10.0	12.0	10.0	17.0
P7	11.0	14.0	11.0	20.0
P8	13.0	14.0	13.0	21.0
Avg WT: 6.625	**Throughput:** 0.381	**Util:** 100%		

SJF Non-Preemptive

Gantt: `0-3[P1] 3-4[P4] 4-6[P3] 6-8[P6] 8-9[P8] 9-12[P7] 12-16[P2] 16-21[P5]`

- **Avg WT:** 3.75
- **Throughput:** 0.381
- **Util:** 100%

Round Robin (q=4.0)

Gantt: `0-3[P1] 3-7[P2] 7-9[P3] 9-10[P4] 10-14[P5] 14-16[P6] 16-19[P7] 19-20[P8] 20-21[P5]`

- **Avg WT:** 7.0
- **Throughput:** 0.381
- **Util:** 100%

Multi-Level Queue (MLQ)

- **Queue 1 (RR q=2):** Avg WT: 7.0
- **Queue 2 (FCFS):** No Data (NaN)
- **Queue 3 (Priority):** Avg WT: 7.75

Scenario D Results

FCFS / SJF / RR (q=4)

Gantt: `0-4[P1] 4-8[P2] 8-12[P3] 12-16[P4] 16-20[P5] 20-24[P6] 24-28[P7] 28-32[P8]`

PID	WT	TAT	RT	CT
P1	0.0	4.0	0.0	4.0
P2	3.0	7.0	3.0	8.0
P3	6.0	10.0	6.0	12.0
P4	9.0	13.0	9.0	16.0
P5	11.0	15.0	11.0	20.0
P6	14.0	18.0	14.0	24.0
P7	17.0	21.0	17.0	28.0
P8	19.0	23.0	19.0	32.0
Avg WT: 9.875	**Throughput:** 0.25	**Util:** 100%		

Priority Non-Preemptive

Variant 1: Avg WT: 9.875

Variant 2: Avg WT: 10.25

Multi-Level Queue (MLQ)

- **Queue 1 (RR q=2):** Avg WT: 8.16
- **Queue 2 (FCFS):** Avg WT: 0.0
- **Queue 3 (Priority):** Avg WT: 16.875

Scenario E Results

FCFS

Gantt: `0-2[P1] 2-4[P2] 4-8[P3] 8-12[P4] 12-18[P5] 18-20[P6] 20-24[P7] 24-26[P8]`

- **Avg WT:** 9.0
- **Throughput:** 0.308
- **Util:** 100%

SJF Non-Preemptive

Gantt: `0-2[P1] 2-4[P2] 4-6[P6] 6-8[P8] 8-12[P3] 12-16[P4] 16-20[P7] 20-26[P5]`

- **Avg WT:** 6.5
- **Throughput:** 0.308
- **Util:** 100%

Priority Non-Preemptive

- **Avg WT:** 9.5 (Both Variants)
- **Throughput:** 0.308

Round Robin (q=4.0)

- **Avg WT:** 9.25
- **Throughput:** 0.308

Multi-Level Queue (MLQ)

- **Queue 1 (RR q=2):** Avg WT: 6.33
- **Queue 2 (FCFS):** Avg WT: 1.0
- **Queue 3 (Priority):** Avg WT: 10.0

3. Performance Comparison Across Algorithms

Scenario A Comparison

Algorithm	Avg WT	Throughput	Utilization
FCFS	12.5	0.235	100%
SJF (Non-Pre)	**7.5**	0.235	100%
Priority (Non-Pre)	9.5 / 9.12	0.235	100%
Round Robin (q=4)	14.875	0.235	100%
MLQ (Queue 1)	6.83	0.272	100%

- **SJF** provided the best Waiting Time among standard algorithms (7.5).
- **Round Robin** suffered from the highest Waiting Time (14.875), indicating the time quantum may have been too large for short processes.
- **MLQ Queue 1** outperformed all standard algorithms, likely due to a smaller time quantum favoring interactive tasks.

Scenario B Comparison

Algorithm	Avg WT	Throughput	Utilization
FCFS	0.25	0.276	68.97%
SJF (Non-Pre)	0.25	0.276	68.97%
Priority (Non-Pre)	0.25	0.276	68.97%
Round Robin (q=4)	0.25	0.276	68.97%
MLQ (Queue 1)	**0.0**	0.207	58.62%

- **Identical Performance:** The presence of significant idle times synchronized the execution flow, neutralizing algorithm differences.
- **Low Utilization:** All algorithms dropped to ~69% utilization due to idle gaps.

Scenario C Comparison

Algorithm	Avg WT	Throughput	Utilization
FCFS	6.625	0.381	100%
SJF (Non-Pre)	**3.75**	0.381	100%
Priority (Non-Pre)	6.625	0.381	100%
Round Robin (q=4)	7.0	0.381	100%

- **SJF Superiority:** SJF drastically reduced waiting time (almost half of FCFS).
- **FCFS & Priority Equality:** Suggests the arrival order happened to match the priority order.

Scenario D Comparison

Algorithm	Avg WT	Throughput	Utilization
FCFS	9.875	0.25	100%
SJF (Non-Pre)	9.875	0.25	100%
Round Robin (q=4)	9.875	0.25	100%

- **Uniformity:** FCFS, SJF, and RR yielded identical results, implying processes arrived in order of burst length (Shortest -> Longest), rendering SJF identical to FCFS.

Scenario E Comparison

Algorithm	Avg WT	Throughput	Utilization
FCFS	9.0	0.308	100%
SJF (Non-Pre)	**6.5**	0.308	100%
Priority (Non-Pre)	9.5	0.308	100%
Round Robin (q=4)	9.25	0.308	100%

- **Trade-off:** SJF optimized waiting time effectively (6.5), while Priority and RR lagged behind FCFS in this specific instance.

Comparative Note: Priority Preemptive Scheduling

Feature	Priority Non-Preemptive (Tested)	Priority Preemptive (Comparative Note)
Core Mechanism	A higher-priority process must wait for the currently running process to complete its CPU burst.	A higher-priority process immediately interrupts a lower-priority process.
Responsiveness	Lower. Critical tasks may experience higher waiting times if a long-running, lower-priority task is executing.	Higher. Guarantees immediate service for urgent, high-priority tasks (e.g., Scenario C or E).
Staggered High-Prio (C, E)	Waiting time is dependent on the running process's remaining time.	Significantly Reduced Waiting Time for critical tasks.
Sparse/Idle (B)	Performance is nearly identical because no preemption conditions arise	Performance is nearly identical because no preemption conditions arise
Sequential (D)	Performance is nearly identical as processes arrive in a favorable order.	Performance is nearly identical as processes arrive in a favorable order.
Overhead	Lower (less context switching).	Higher (due to frequent context switching for interruptions).

Implementation Design Clarification: Multi-Level Queue (MLQ):

- **Independent Queue Evaluation:** Each queue (e.g., Queue 1: RR, Queue 2: FCFS, Queue 3: Priority) is evaluated as an **independent entity**.
- **Isolation of Scheduling Logic:** Each queue possesses its **own internal scheduling logic** (RR, FCFS, or Priority).
- **Purpose:** This design choice **isolates the queues** to demonstrate how different algorithms handle specific, segregated subsets of processes.
- **Multi-Level Behavior Highlighted:** This approach effectively highlights the intended multi-level behavior, where **distinct classes of processes** (e.g., interactive vs. batch) are segregated and managed by the algorithm best suited for their specific workload characteristics, rather than competing on a single unified CPU timeline.

Global Summary

Across all scenarios, **SJF (Non-Preemptive)** consistently demonstrated the lowest Average Waiting Time when process bursts varied. **Round Robin** generally incurred higher waiting times due to context switching overhead but maintained fairness. **MLQ** showed high variance; Queue 1 (RR q=2) often performed exceptionally well, while Queue 3 (Priority) often suffered from starvation or high delays.

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4. Test Cases and Sample Runs

The following data summarizes the workload characteristics derived from the simulation logs.

Scenario A

- **Characteristics:** Continuous load, no idle time.
- **Process count:** 8 (P1-P8).
- **Burst estimates:** P1(5), P2(3), P3(8), P4(6), P5(2), P6(4), P7(1), P8(5).

Scenario B

- **Characteristics:** Sparse load with frequent idle gaps.
- **Idle Times:** 3.0-5.0, 10.0-11.0, 18.0-20.0, 21.0-25.0.
- **Burst estimates:** P1(3), P2(4), P3(1), P4(2), P5(3), P6(2), P7(1), P8(4).

Scenario C

- **Characteristics:** Medium load, varying bursts.
- **Burst estimates:** P1(3), P2(4), P3(2), P4(1), P5(5), P6(2), P7(3), P8(1).

Scenario D

- **Characteristics:** Sequential load where Arrival Order roughly matches Burst Size order.
- **Burst estimates:** P1(4), P2(4), P3(4), P4(4), P5(4), P6(4), P7(4), P8(4) (Uniform bursts).

Scenario E

- **Characteristics:** Mixed bursts.
- **Burst estimates:** P1(2), P2(2), P3(4), P4(4), P5(6), P6(2), P7(4), P8(2).

5. Conclusion

- **FCFS** is simple but slow under load due to the convoy effect. **SJF Non-Preemptive** minimizes average waiting time but is unfair and requires impossible future knowledge. **Non-Preemptive Priority** respects hierarchy but risks starvation; **Preemptive Priority** improves responsiveness but increases overhead. **Round Robin** provides excellent fairness and response time but suffers in average waiting time if the quantum is poor. **MLQ** offers rigid process isolation, while **MLFQ** complex but adaptive best balances responsiveness and fairness. No algorithm is "perfect" the optimal choice depends on matching the strategy to the system's specific demands for responsiveness and throughput.

Advantages and Disadvantages of CPU Scheduling Algorithms

Algorithm	Advantages	Disadvantages
First-Come, First-Served (FCFS)	Implementation simplicity.	Suffers significantly from the convoy effect under heavy loads (high Average Waiting Time).
Shortest Job First (SJF) Non-Preemptive	Guarantees the mathematical minimum for average waiting time (lowest AWT).	Introduces fairness issues; requires future knowledge of burst times (impractical).
Priority Non-Preemptive	Ensures hierarchy is respected.	Risks indefinite starvation for low-priority jobs (high AWT variance); lower responsiveness.
Round Robin (RR)	Excels at fairness and response time.	Struggles with average waiting time if the time quantum is unoptimized; higher context switching overhead.
Multi-Level Queue (MLQ)	Provides isolation for different process classes/workloads.	Remains rigid; high variance in performance (some queues suffer high delays/starvation).
Priority Preemptive	Improves responsiveness for critical tasks (low response time).	Introduces higher context switching overhead.
Multi-Level Feedback Queue (MLFQ)	Best balance of responsiveness and fairness; adapts to process behavior dynamically.	Most complex to implement.